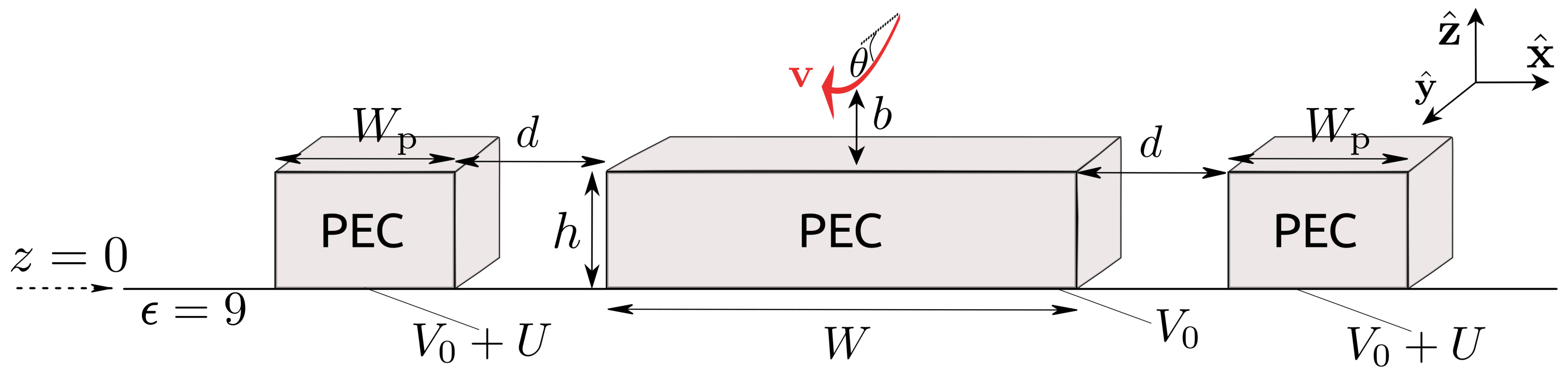
# Electron-coupling to waveguide: code of the EELS

$$\Gamma(x,\omega) \approx 2\int_{\underbrace{z_{\min}}_{\text{step 3b}}}^{\infty} \frac{v\,\mathrm{d}z}{\sqrt{|2e\underbrace{(a_1 z + b_1)}_{\text{approximation of the potential}}/m_e\gamma_e - c^2\beta^2\sin^2\theta|}}\underbrace{\frac{\mathrm{d}\Gamma(\omega,x,z)}{\mathrm{d}y}}_{\text{step 1}}.$$

integration in step 4




1) run_EELS_rect_wg_along_z : obtain dEELS(z)/dy [1/eV*nm] for many energies

2) split_positive_eels.sh : split EELS in 1 file per 1 energy and retain only EELS >0

3) run_potential_approx_and_find_bmin.sh :

a) Run e.out to create normalized potential data approximated to a linear function!! fit and save a1 and b1 around zmin +/- delta

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g++ -o e.out e.cpp
```

files e.cpp and e.h

b) Run Python script to obtain bmin(V0,theta)
find_bmin_vs_V0_theta.py

$$\frac{z_{\min}}{W} = \frac{1}{a_1}\left(\frac{m_e c^2\gamma_e}{2e}\frac{\beta^2\sin^2\theta}{V_0} - b_1\right), \longrightarrow \theta,\ V_0 \text{ for a fixed } z_{\min}$$

4) EELS_integrated_over_z.h and EELS_integrated_over_z.cpp: save "energy | Pintegrated [1/eV]" for each solution V0, theta

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g++ -o EELS_integrated_over_z.out EELS_integrated_over_z.cpp
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5) calculate_total_EELS_vs_theta.py: save "energy | EELS tot" for each solution V0, theta where EELS tot is the EELS integrated over z (data from step 4) and over the mode. Integration over the mode by fitting the data with a Lorentzian
fit_peaks.py